# Practice workbook · How hard drives, optical discs and SSDs store data

How data is physically stored and read on magnetic, optical and solid state devices, and reasons to choose a solid state drive over a hard disc.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: CSC-T4-07

# 1. Question 2 · part b.ii · 3 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part b.ii only.

**2** Computer systems use hardware and software.

(a) Two types of software are application software and system software.

(i) State the purpose of application software.

**(1)**

.....

.....

(ii) One function of an operating system is to provide a user interface.

State **three** other functions of an operating system.

**(3)**

1 .....

.....

2 .....

.....

3 .....

.....

OLevels.com SCHOOL

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

# 1. Question 2 · part b.ii · 3 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) Computers use different types of storage.

(i) Explain **one** reason why computers need both RAM and ROM.

**(2)**

(ii) Describe how data is read from a magnetic hard drive.

**(3)**

OLevels.COM SCHOOL

(c) Computers perform the fetch-decode-execute cycle.

Complete the table by writing the numbers 1 to 4 in the sequence number column to put each stage in the correct order.

| Stage | Sequence number |
|---|---|
| The instruction is placed onto the data bus | |
| The Arithmetic and Logic Unit (ALU) performs a calculation | |
| The address of the next instruction is placed onto the address bus | |
| The instruction is decoded by the control unit | |

**(4)**

## 1. Question 2 · part b.ii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

## 2. Question 1 · part e · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part e only.

**1** Computers are made up of both hardware and software components.

(a) Computers use a three-step computational model.

Complete the diagram with the names of the **three** steps.

**(3)**



(b) Identify the hardware component responsible for controlling the fetch-decode-execute cycle.

**(1)**

- ☒ **A** Cache
- ☒ **B** Central processing unit
- ☒ **C** ROM
- ☒ **D** Address bus

(c) Von Neumann developed the stored program concept that permits two different types of item to reside in memory.

Name these **two** types.

**(2)**

1 .....

.....

2 .....

.....

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

## 2. Question 1 · part e · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part e only.

(d) Some computers use virtual memory.

Explain how virtual memory works.

(2)

(e) An optical storage device writes data onto a CD or DVD.

Describe how data is stored physically on optical media.

(2)

(f) A disc-formatting program is an example of a type of software.

Identify the type.

(1)

- ☒ **A** Security software
- ☒ **B** Application software
- ☒ **C** Translation software
- ☒ **D** System software

**(Total for Question 1 = 11 marks)**

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

## 2. Question 1 · part e · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

# 3. Question 3 · part b.i · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part b.i only.

**3** Zakir wants to buy a new laptop.

(a) He knows that a laptop has both hardware and software.

Explain why the laptop needs both hardware and software.

**(2)**

(b) Zakir needs to choose between a solid state drive and a hard disc drive.

(i) Give **two** reasons, other than cost and capacity, why Zakir may choose a solid state drive rather than a hard disc drive.

**(2)**

1

2

(ii) Solid state drives store data on flash memory, which is similar to RAM.

Explain **one** difference between flash memory and RAM.

**(2)**

(c) One of the laptops has 4 megabytes (MB) of cache memory.

Explain the purpose of cache memory.

**(2)**

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

## 3. Question 3 · part b.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

# 4. Question 5 · part d.i · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part d.i only.

**5** Viza Health Centre is located in the North East of England.

(a) The health centre uses artificial intelligence to provide a symptom-checking service for its patients.

Patients log on to the website and input their symptoms.

(i) Describe how artificial intelligence could identify what is wrong with them.

**(2)**

(ii) Give **one** reason why a patient may not want to use this online service.

**(1)**

(b) The health centre has clinics in two buildings: Cleveland and Stockton.

The network server is in the Cleveland building.

(i) Name the type of network used to access the server from within the Cleveland building.

**(1)**

(ii) Name the type of network used to access the server from the Stockton building.

**(1)**

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

# 4. Question 5 · part d.i · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part d.i only.

(c) The network is at risk of an eavesdropping attack.

Identify the description of eavesdropping.

(1)

- ☒ **A** Tricking people into giving information by sending emails pretending to be from someone in authority
- ☒ **B** Spying on someone using a computer
- ☒ **C** Intercepting information as it is transmitted over a network
- ☒ **D** Redirecting a user from a genuine website to a fake one

(d) Doctors use laptops when they visit patients in their homes.

(i) The laptops have solid state drives.

Explain **one** reason why a solid state drive is better than a magnetic hard drive for the laptops.

(2)

OLevels.com SCHOOL

(ii) Describe how data is stored on a solid state drive.

(2)

(iii) The laptops have two types of memory.

Complete the table by adding **one** tick (✓) to match **each** description to the type of memory used.

(2)

| Description | RAM | ROM |
|---|---|---|
| Stores the boot up sequence | | |
| The contents are lost when the laptop is shut down | | |

**(Total for Question 5 = 12 marks)**

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

## 4. Question 5 · part d.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

# 5. Question 5 · part d.ii · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part d.ii only.

**5** Viza Health Centre is located in the North East of England.

(a) The health centre uses artificial intelligence to provide a symptom-checking service for its patients.

Patients log on to the website and input their symptoms.

(i) Describe how artificial intelligence could identify what is wrong with them.

**(2)**

(ii) Give **one** reason why a patient may not want to use this online service.

**(1)**

(b) The health centre has clinics in two buildings: Cleveland and Stockton.

The network server is in the Cleveland building.

(i) Name the type of network used to access the server from within the Cleveland building.

**(1)**

(ii) Name the type of network used to access the server from the Stockton building.

**(1)**

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

# 5. Question 5 · part d.ii · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part d.ii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(c) The network is at risk of an eavesdropping attack.

Identify the description of eavesdropping.

(1)

- ☒ **A** Tricking people into giving information by sending emails pretending to be from someone in authority
- ☒ **B** Spying on someone using a computer
- ☒ **C** Intercepting information as it is transmitted over a network
- ☒ **D** Redirecting a user from a genuine website to a fake one

(d) Doctors use laptops when they visit patients in their homes.

(i) The laptops have solid state drives.

Explain **one** reason why a solid state drive is better than a magnetic hard drive for the laptops.

(2)

OLevels.com SCHOOL

(ii) Describe how data is stored on a solid state drive.

(2)

(iii) The laptops have two types of memory.

Complete the table by adding **one** tick (✓) to match **each** description to the type of memory used.

(2)

| Description | RAM | ROM |
|---|---|---|
| Stores the boot up sequence | | |
| The contents are lost when the laptop is shut down | | |

**(Total for Question 5 = 12 marks)**

## 5. Question 5 · part d.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL